1 Programming, Data Structures and Algorithms

1.1 Programming

1. Programming in Python

1.2 Data Structures

1. Stack
2. Queue
3. Linked List
4. Trees
5. Hash Tables

1.3 Searching Algorithms

1. Linear Search
2. Binary Search

1.4 Sorting Algorithms

1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort

1.5 Graph Theory and Algorithms

1. Introduction to Graph Theory
2. Graph Traversals
3. Shortest Path Algorithms